

FR-016 M1 — Multi-Agent Orchestration Contracts

FR016 — `sources/optional/m1.md`

Masterclass source package snapshot

Functional Requirement	FR-016
Edition	Engineering Learning Academy — Source Package snapshot
Status	Complete (M1) — contracts frozen; succeeded by M2 runtime
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via `python scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

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Unit

FR-016 M1 — Multi-Agent Orchestration Contracts

Date: 2026-08-06

Status: Complete (M1) — contracts frozen; succeeded by M2 runtime

Architecture: [ADR-008](#) (Accepted at M1)

Preceding: [M0 engineering spike](#) (Accepted with revisions)

Succeeded by: [M2 supervisor runtime / go-no-go](#); FR-016 later **Complete / Frozen** — [acceptance](#)

Does not begin (historical M1): M2 runtime was deferred to the next milestone

1. Architectural decisions

Decision	Choice	Why
Package	<code>career_intelligence.multi_agent</code>	Distinct from FR-008 <code>orchestration</code> and frozen FR-015 <code>agent</code>
Topology	DOS + BOPA + OBS	Owner-accepted narrow M0
BOPA	Unchanged; referenced via registry	ADR-007 freeze; no theatre split
OBS	Strictly read-only briefing	Distinct ToolPolicy; value BOPA must not absorb
Supervisor	Delegation only	No domain super-user
Handoffs	Typed, append-only, idempotent	No free-form chat authority
Policies	DelegationPolicy + per-specialist ToolPolicy	Privilege escalation fail-closed
LLM	Optional propose only (M2+)	Deterministic default
M1 deliverable	Contracts + ADR-008 + unit tests	No runtime / CLI / adapters

```
Owner OrchestrationGoal
  → OrchestrationObservation (derived)
  → BriefingNeedClass / approved specialists
  → SpecialistDelegationProposal (M2+ proposer; M1 data)
  → evaluate_delegation_policy
  → Handoff (append-only)
  → specialist ToolPolicy (OBS or BOPA)
  → (M2+) child run / OperationalBrief
  → orchestration stop reason
```

2. Why OBS (value beyond BOPA)

OBS exists only where a **read-only briefing delta** would otherwise force broadening BOPA's mutating `prepare_for_owner_review` responsibility.

BriefingNeedClass	Observable signal	Why not BOPA alone
<code>pipeline_advises_against_preparation</code>	Pipeline in submitted / interviewing / offer / terminal	BOPA may still legally prepare on <code>decision=apply</code> ; owner needs brief-first without mutate tools in-policy
<code>cross_surface_ambiguity</code>	Contradictory flags or decision holes with artefacts	Diagnose without preparation authority
<code>truth_blockers_need_synthesis</code>	Owner-facing blocker labels present	Cross-surface brief citing truth + package + prior runs
<code>prior_agent_history_material</code>	Prior AgentRun ids present	Synthesise history without re-running prep
<code>owner_requested_brief_only</code>	Goal <code>brief_opportunity_readiness</code> / <code>brief_only</code>	Mutate must stay impossible under OBS policy
<code>batch_triage</code> / <code>post_specialist_synthesis</code> / <code>brief_before_mutate</code>	Reserved for M2+ orchestration modes	Same isolation rationale
<code>no_briefing_delta</code>	Assessed apply + missing package, no special signals	Prefer direct BOPA; OBS not required

Theatre rejection (binding): Prep Specialist / Truth Specialist / Review Specialist personas that wrap FR-010/011/014 or BOPA actions are **out of scope**.

3. Specialist boundaries

3.1 DOS (supervisor)

Owns	Must not
Observe orchestration state	Call mutating domain services
Evaluate DelegationPolicy	Bypass specialist ToolPolicy
Create/validate typed handoffs	Waive truth / submit / advance pipeline
Enforce global budgets / loop controls	Inherit specialist tools
Aggregate child stops into orchestration stop	Free-form agent chat

3.2 OBS allow-list

`inspect_readiness`, `inspect_pipeline_context`, `inspect_truth_blockers`, `inspect_agent_history`, `compose_brief`, `recommend_delegation`, `request_owner_review`, `stop`.

`recommend_delegation` records a suggestion only — **DelegationPolicy** still admits any actual specialist invocation.

3.3 BOPA allow-list

Unchanged FR-015: `inspect_readiness`, `run_preparation`, `verify_package`, `validate_truth_package`, `request_owner_review`, `stop`.

Referenced via `BOPA_SPECIALIST` in the registry; not copied as a parallel mutate surface.

4. Implementation summary

API	Role
<code>OrchestrationGoal</code> / <code>OrchestrationRun</code> / <code>OrchestrationAuditEvent</code>	Parent run + append-only audit shapes
<code>OrchestrationObservation</code>	Derived cross-surface observation
<code>Handoff</code>	Typed handoff; supervisor-sourced; idempotency key
<code>OperationalBrief</code>	OBS output (derived)
<code>SpecialistDelegationProposal</code> / <code>DelegationDecision</code>	Delegation proposal + policy result
<code>evaluate_delegation_policy</code>	Deterministic DelegationPolicy
<code>ObsActionProposal</code> / <code>evaluate_obs_action_policy</code>	OBS ToolPolicy
<code>classify_briefing_needs</code> / <code>obs_adds_value_beyond_bopa</code>	OBS justification helpers
<code>SPECIALIST_REGISTRY</code>	Static BOPA + OBS contracts
<code>new_orchestration_run_id</code> / <code>new_handoff_id</code> / ...	<code>orr_</code> / <code>hof_</code> / <code>oe_</code> / <code>obr_</code> ids

Public surface: `career_intelligence.multi_agent`.

M1 does **not** implement DOS runtime, OBS/BOPA executors, proposers, stores, CLI, or frameworks.

5. Validation results

Unit

`tests/unit/multi_agent/` — **32 passed**.

Check	Result
Id patterns / generators	Pass
Extra fields forbidden	Pass
BOPA allow-list reference unchanged	Pass
OBS cannot mutate invariant	Pass
Briefing-need classification (OBS value)	Pass
DelegationPolicy allow/deny / limits / circular / repeat	Pass
Brief goal cannot delegate BOPA	Pass
OBS ToolPolicy allow/deny / no-progress	Pass
Handoff supervisor-source + accept rules	Pass
No privilege escalation via recommend_delegation	Pass

```
python -m pytest tests/unit/multi_agent/ -q
32 passed
```

6. Documentation updated

Document	Change
ADR-008	Accepted — DOS + BOPA + OBS; theatre rejected; M2 go/no-go
M0 spike	Marked Accepted with revisions
Functional specification	FR-016 M1 contracts
Domain model	Multi-agent entities
ADR index / changelog / roadmap pointers	M1 status

7. M2 go/no-go gate (mandatory)

M2 may implement runtime **only** to gather evidence. Before M3:

Question	Evidence required
Does DOS add value beyond <code>cic agent</code> / reporting?	Side-by-side owner journeys
Does OBS remove a meaningful owner task?	Interviewing / truth-blocked / history cases
Do handoffs + separated permissions improve safety/audit/extensibility?	Illegal delegation / injection / audit replay
Is complexity proportionate?	Honest product assessment
Continue / learning-proof / defer to Job Discovery?	Explicit owner decision

8. Non-goals (M1)

- No production supervisor loop
 - No OBS/BOPA adapter execution from DOS
 - No CLI namespace
 - No framework integration
 - No job discovery, submit, pipeline mutation, truth waiver
 - No Prep/Truth/Review persona agents
 - No changes to frozen BOPA behaviour
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9. M1 acceptance

M1 contracts + ADR-008 are complete. Proceed to M2 only on owner request, with the go/no-go review as a hard gate before M3.